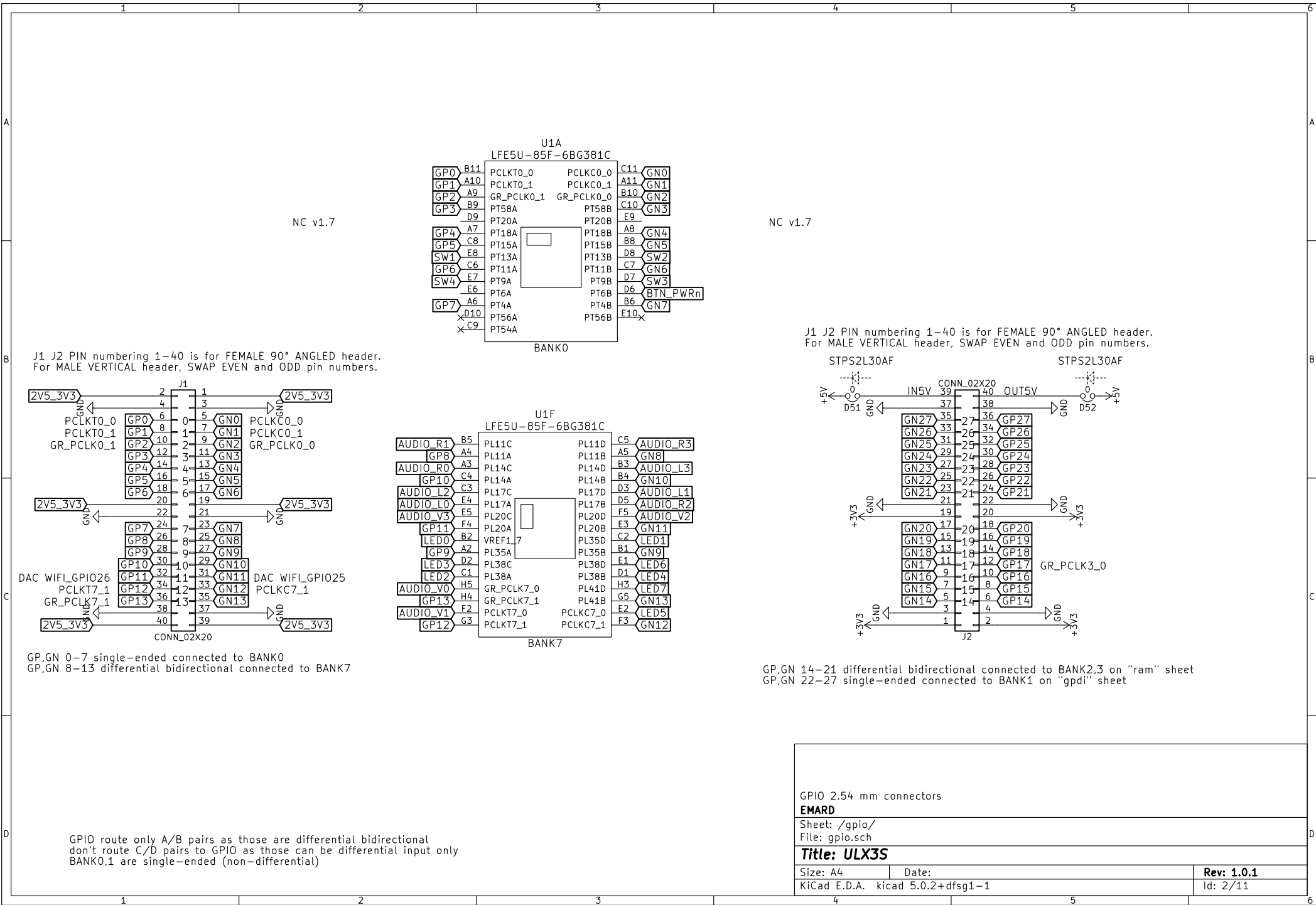


Sheet: /
File: ulx3s.sch

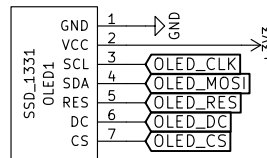
Title: ULX3S

Size: A4	Date:
KiCad E.D.A. kicad 5.0.2+dfsg1-1	

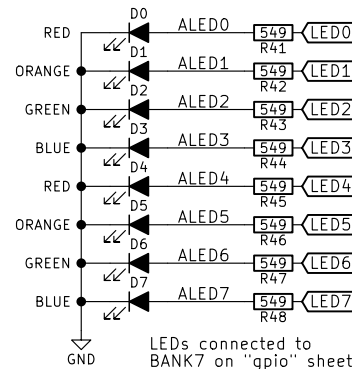
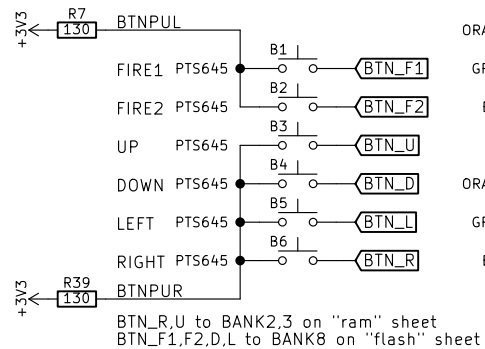
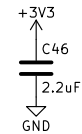
Rev: 3.0.8
Id: 1/11



SSD1306 B/W or SSD1331 COLOR
compatible OLED 0.96" or 1.3" PCB
14x14 units
1 unit = 2.54 mm

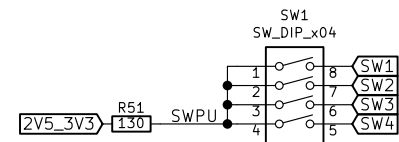


OLED connected to
BANK6 on "usb" sheet



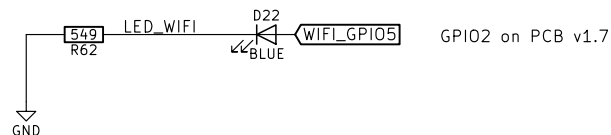
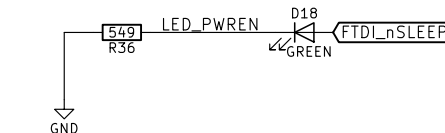
LEDs connected to
BANK7 on "gpio" sheet

SW1 heeft geen relevantie met bootmode



DIP switch connected to
BANK0 on 'gpio' sheet

To fix issues with FT231XS rev A,B,C
Short-circuit D18 LED, but then
board cannot keep awake by USB.
chip rev D works properly
See TN140_FT231X Errata



Buttons, LEDs, OLED display

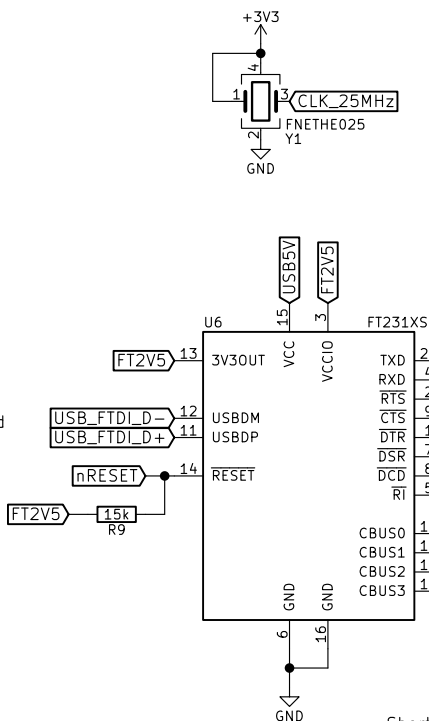
EMARD

Sheet: /blinky/
File: blinky.sch

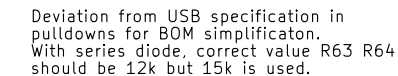
Title: ULX3S

Size: A4 Date:
KiCad E.D.A. kicad 5.0.2+dfsg1-1

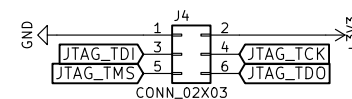
Rev: 1.0.0
Id: 4/11



USB pull lines connected to
BANK1 on "qpdi" sheet



warning:
ULX3S has different pinout
for simpler PCB routing and
because FT230X has weak CTS
drive capability. (Undocumented,
FLEAfpga mail from 13-Nov-2015)
ULX2S pinout was:
TCK = DSR
TMS = RI
TDI = CTS
TDO = DCD



Id: 6/11

SD card connected according to
 Arduino->Examples->ESP32->SD_MMC_Test
https://github.com/espressif/esp-idf/blob/master/examples/storage/sd_card/README.md

GPIO12 probably should be held low at reset or disconnected
 other option is to burn flash voltage efuse for 3.3V operation
 for GPIO2 and GPIO12 bootstrapping considerations, read
https://github.com/espressif/esp-idf/tree/master/examples/storage/sd_card

GPIO25,26 are DAC channels that can directly replay
 audio material over i2s bus

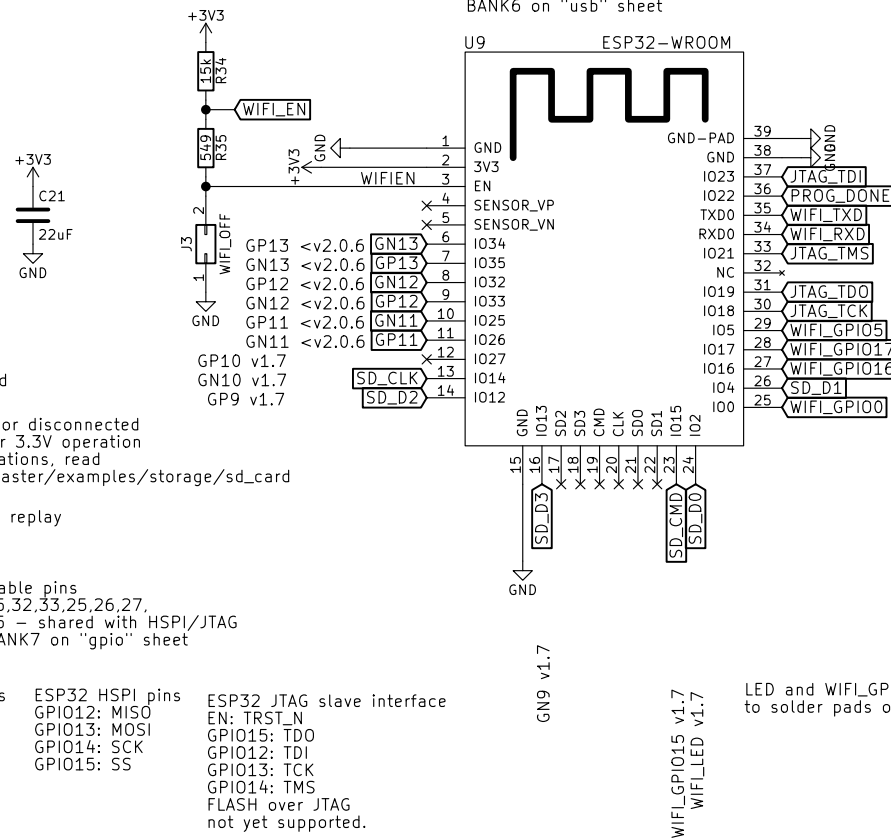
ESP32 ADC capable pins
 GPIO36,39,34,35,32,33,25,26,27,
 GPIO14,12,13,15 — shared with HSPI/JTAG
 Connected to BANK7 on "gpio" sheet

ESP32 VSPI pins
 GPIO5: SS
 GPIO18: SCK
 GPIO19: MISO
 GPIO23: MOSI

ESP32 HSPI pins
 GPIO12: MISO
 GPIO13: MOSI
 GPIO14: SCK
 GPIO15: SS

ESP32 JTAG slave interface
 EN: TRST_N
 GPIO15: TDO
 GPIO12: TDI
 GPIO13: TCK
 GPIO14: TMS
 FLASH over JTAG
 not yet supported.

WiFi and SDcard connected to
 BANK6 on "usb" sheet



PROG_DONE on
 "flash" sheet

Programming pins:
 TXD RXD EN GPIO0

NC v1.7, LED v1.8
 NC v1.7

NC v1.7

Strapping pins
 (avoid using to avoid
 boot problems)
 GPIO0,2,4,5,12,15

LED and WIFI_GPIO17
 to solder pads of PCB

WiFi module ESP32

EMARD

Sheet: /wifi/

File: wifi.sch

Title: ULX3S

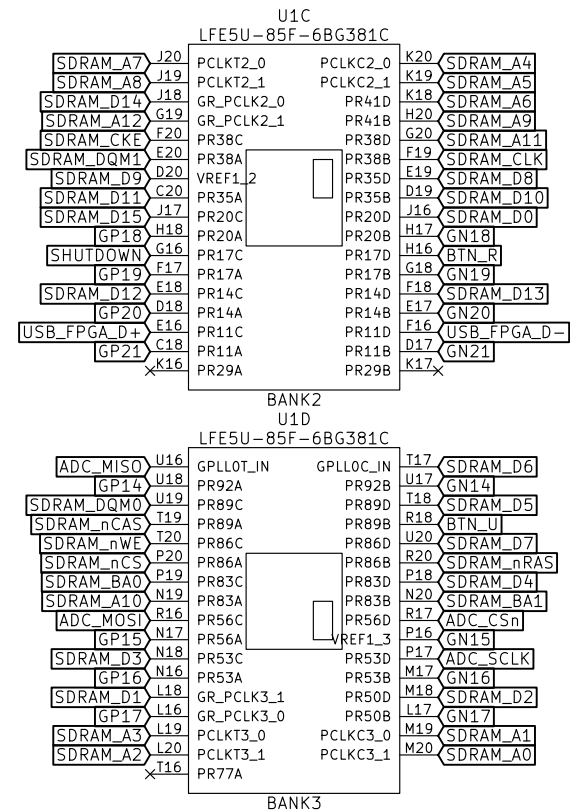
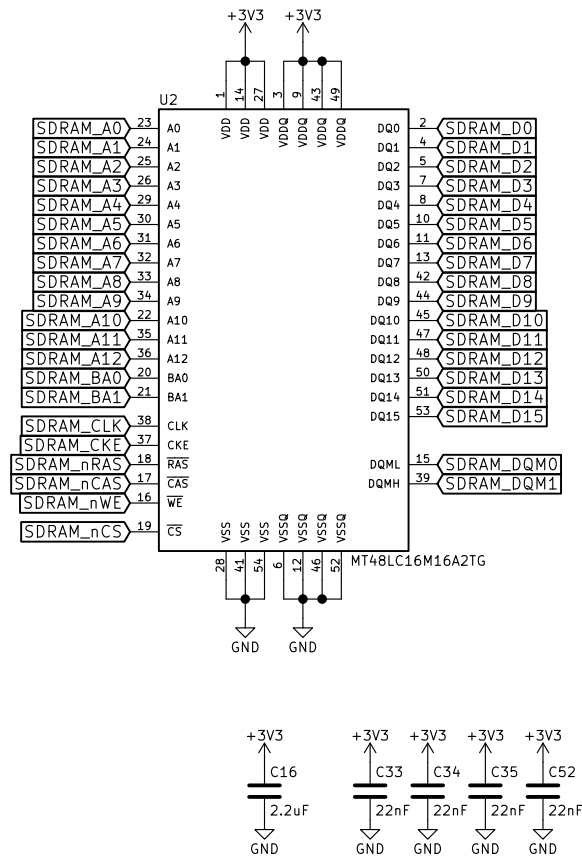
Size: A4

Date:

KiCad E.D.A. kicad 5.0.2+dfsg1-1

Rev: 1.0.2

Id: 7/11



SDRAM memory

EMARD

Sheet: /ram/

File: ram.sch

Title: ULX3S

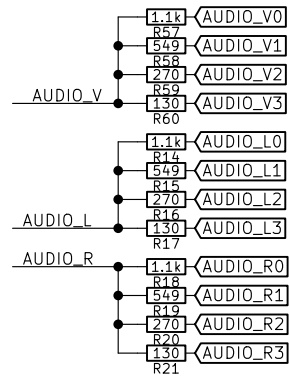
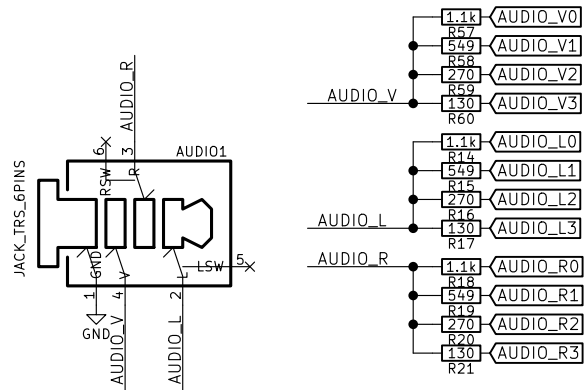
Size: A4

Date:

KiCad E.D.A. kicad 5.0.2+dfsg1-1

Rev: 1.0.0

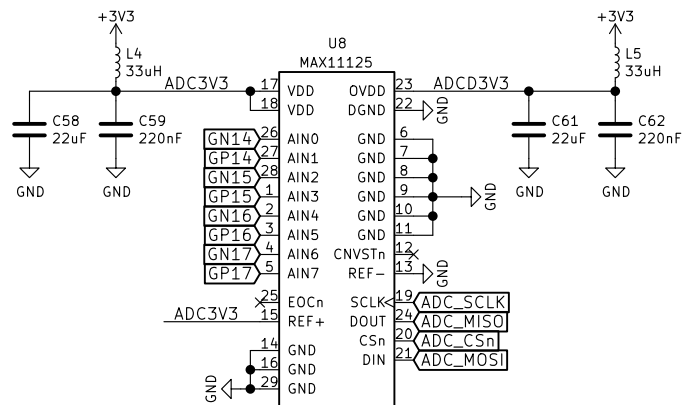
Id: 8/11



JACK pinout for SJ-43516-SMT-TR
<http://www.cui.com/product/resource/sj-4351x-smt-series.pdf>
 pin 1 - sleeve (GND)
 pin 2 - tip (left channel)
 pin 3 - ring1 (right channel)
 pin 4 - ring2 (video)
 pin 5 - tip switch
 pin 6 - ring1 switch

Audio connected to
 BANK7 on "gpio" sheet

Output resistance: 75 ohm
 Internal resistance of FPGA pin: 10 ohm
 $1/(1/(130+10)+1/(270+10)+1/(549+10)+1/(1100+10))=74.6$



ADC SPI connected to
 BANK3 of "ram" sheet

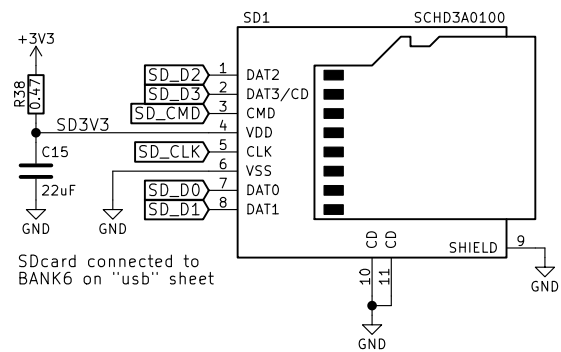
Analog audio and video
EMARD

Sheet: /analog/
 File: analog.sch

Title: ULX3S

Size: A4 Date:
 KiCad E.D.A. kicad 5.0.2+dfsg1-1

Rev: 1.0.4
 Id: 9/11



minimum pins for compatible mode
SD_CLK, SD_CMD, SD_D0, SD_D3

SD card
EMARD

Sheet: /sdcard/
File: sdcard.sch

Title: ULX3S

Size: A4 Date:
KiCad E.D.A. kicad 5.0.2+dfsg1-1

Rev: 1.0.0
Id: 10/11

